



USL Assignment 7

K-Means Clustering for Customer Segmentation

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MDS THEORY

1. Introduction to Multidimensional Scaling (MDS)

- K-Means Clustering is an unsupervised machine learning algorithm used to group data points into a predefined number of clusters based on similarity.
- It partitions the dataset into **K clusters**, where each data point belongs to the cluster with the nearest mean (centroid).
- Unlike supervised learning, K-Means does not require labeled data and focuses only on the inherent structure of the dataset.



2. Working of K-Means Algorithm

The algorithm works iteratively through the following steps:

1. Select the number of clusters (K)
2. Initialize K centroids randomly
3. Assign each data point to the nearest centroid
4. Recalculate centroids as the mean of assigned points
5. Repeat steps 3 and 4 until convergence

Convergence occurs when cluster assignments no longer change.

3. Role of Centroids

Centroids represent the center of each cluster.

- They are calculated as the mean of all points in a cluster
- They are updated in each iteration
- Final centroids determine cluster boundaries

4. WCSS (Within-Cluster Sum of Squares)

WCSS is a measure of the compactness of clusters.

$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$



Where:

- C_i = cluster
- μ_i = centroid

Lower WCSS indicates better clustering.

5. Elbow Method

The Elbow Method is used to determine the optimal number of clusters.

Steps:

- Run K-Means for different values of K
- Plot K vs WCSS

Observation:

- WCSS decreases as K increases
- The point where the decrease becomes less significant forms an “elbow”

This point represents the optimal number of clusters.

6. Silhouette Score

Silhouette Score measures how well each data point fits within its cluster.



$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Where:

- a = intra-cluster distance
- b = nearest-cluster distance

Interpretation:

- Close to +1 → well-clustered
- Around 0 → overlapping clusters
- Negative → incorrect clustering

7. Advantages of K-Means

- Simple and easy to implement
- Efficient for small to medium datasets
- Works well for clearly separable clusters

8. Limitations of K-Means

- Requires predefined number of clusters
- Sensitive to outliers



- Assumes spherical cluster shapes
- Performance depends on initialization